USN [ ][ ][ ][ ][ ][ ][ ][ ][ ][ ] **[23TCSE304]**

THIRD SEMESTER B.Tech EXAMINATION, NOVEMBER-2024

DATABASE MANAGEMENT SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: **i.** Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART-A			**5 x2=10**
1.	**a.**	What are the main actors on the scene in a database environment?	L1	CO1	**02**
	b.	What are some common relational model constraints?	L1	CO2	**02**
	c.	Name three common data types used in SQL for defining table columns.	L1	CO3	**02**
	d.	What is a view in SQL?	L1	CO4	**02**
	e.	What are the four main desirable properties of a transaction in database systems?	L1	CO5	**02**
		PART-B			**6 X 5=30**
2.	**a.**	How can the concept of weak entities be presented in an ER diagram?	L3	CO1	**05**
		OR			
	b.	Given a database schema, what should be done to ensure physical data independence when adding new storage structures?	L3	CO1	**05**
3.	**a.**	Design a relational database for a library system. Apply the ER-to-Relational mapping rules to convert an entity "Book" with attributes ISBN, Title, and Author into a relational schema.	L3	CO2	**05**
		OR			
	b.	How does the SELECT operation in relational algebra differ from the PROJECT operation?	L2	CO2	**05**
4.	**a.**	How do aggregate functions like SUM() and COUNT() work in SQL?	L2	CO3	**05**
		OR			
	b.	Write an SQL query to update the salary of an employee with EmployeeID = 101 to 75000. Write an SQL query to create a table named Students with columns StudentID (integer, primary key), Name (varchar), Age (integer), and Gender (varchar).	L3	CO3	**05**
5.	**a.**	Write an SQL query that uses a view to display employee names and their total sales from an Employees and Sales table. The view should be named EmployeeSalesView.	L3	CO4	**05**
		OR			
	b.	Write a stored procedure in SQL that calculates the average salary of employees in a given department and returns it. The department name is passed as an input parameter.	L3	CO4	**05**

6.	**a.**	Given the following relation, apply normalization to 2NF StudentID | CourseID | StudentName | CourseName | Instructor 1 | 101 | P | Math | Mr. A 2 | 102 | Q | History | Ms. B 1 | 102 | R | History | Ms. B	L3	CO5	**05**
		OR			
	b.	Write an SQL query to ensure that transactions on a bank account are atomic using START TRANSACTION, COMMIT, and ROLLBACK.	L3	CO5	**05**
7.	**a.**	Explain the difference between INNER JOIN and LEFT JOIN in SQL. Write an example query for each using tables Employees and Departments.	L4	CO5	**05**
		OR			
	b.	How do stored procedures differ from SQL functions?	L4	CO4	**05**
		PART-C		**6 X 10=60**	
8.	**a.**	Design a three-schema architecture for an online bookstore database system. Describe the external, conceptual, and physical schemas.	L6	CO1	**05**
		OR			
	b.	What is the role of schema architecture in ensuring data independence in database systems.	L6	CO1	**05**
9.	**a.**	What are the advantages and disadvantages of mapping an ER diagram to a relational database schema compared to using a NoSQL database model.	L6	CO2	**05**
		OR			
	b.	What is the role of relational algebra in query optimization within relational database management systems (RDBMS).	L6	CO2	**05**
10.	**a.**	Write an SQL query to create a table called Students with the following fields: StudentID, Name, Age, and Gender. Add a primary key on StudentID and a constraint to ensure Age is between 18 and 30. Find the number of females who are 18 years old and the number of males who are 21 years old	L5	CO3	**05**
		OR			
	b.	Create a table called products that shows the id, name, price and category. Create another table called sales that shows the quantity of each product sold and the date on which each product is sold Find the average sales and total sales of each product, average sales per day and count the number of transactions per day. Also display the product that has the highest total sales	L5	CO3	**05**
11.	**a.**	Evaluate the advantages and disadvantages of using views in SQL for database security and performance.	L6	CO4	**05**
		OR			

	b.	Evaluate the use of cursors in SQL. When would it be appropriate to use cursors, and what are the drawbacks of using them? Give an example	L6	CO4	**05**
12.	**a.**	Create a trigger in SQL that automatically logs any changes made to the Enrollments table into an AuditLog table.	L5	CO5	**05**
		OR			
	b.	Evaluate the importance of serializability in transaction scheduling. Can a schedule be recoverable but not serializable? Explain.	L6	CO5	**05**
13.	**a.**	Evaluate the impact of multivalued dependencies on database design and explain when Fourth Normal Form (4NF) is necessary.	L6	CO5	**05**
		OR			
	b.	Analyze the benefits and potential drawbacks of using triggers in a database system.	L4	CO4	**05**